

AI Decision Recommendation Bot for Student Decision-Making

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Abstract

Students routinely face high-impact academic and career decisions such as selecting a course of study, final-year project topic, internship, or postgraduate program often under uncertainty and conflicting advice. Traditional decision-making approaches among students are largely intuitive, emotion-driven and unstructured, leading to suboptimal outcomes and post-decision regret. This project presents an AI Decision Recommendation Bot, implemented as a Telegram chatbot, that operationalizes a structured, explainable and weighted multi-criteria decision-making framework. The system guides users through problem validation, criteria definition, priority weighting, option scoring, objective computation and transparent explanation of results. Emphasis is placed on robust input validation, edge-case handling and explainable recommendations. The solution demonstrates how conversational AI can transform vague decision problems into rational, defensible outcomes suitable for student contexts.

Keywords

Decision Support System, Telegram Bot, Artificial Intelligence, Weighted Scoring Model, Explainable AI, Student Decision-Making

Introduction

Decision-making is a fundamental cognitive process that significantly influences academic performance, career trajectory and long-term life outcomes. For students, decisions such as choosing a major, selecting a project topic, or accepting an internship often involve multiple competing criteria, including difficulty, career relevance, feasibility, time constraints and personal interest.

Despite the complexity of these decisions, many students rely on intuition, peer influence, or incomplete information. This often results in:

- Overemphasis on a single factor (e.g., financial reward)
- Neglect of long-term consequences
- Poor comparison between alternatives
- Regret after decision implementation

The absence of structured decision-support tools tailored to student needs motivates the development of the AI Decision Recommendation Bot. This project leverages conversational AI and decision theory to guide students through a rational evaluation process using a familiar messaging platform.

Problem Statement

The core problem addressed in this project is the lack of a structured, objective and explainable decision-making framework for students. Existing approaches suffer from the following limitations:

1. Decisions are often emotionally driven rather than evidence-based.
2. Students struggle to define and prioritize decision criteria.
3. There is no systematic way to compare multiple options fairly.

4. Most advice is qualitative, subjective and non-reproducible.
5. Students lack feedback on *why* one option is better than another.

This project seeks to mitigate these issues by embedding a formal decision model within an AI-powered conversational interface.

Objectives of the Project

The primary objectives of this project are:

1. To design a Telegram-based AI bot capable of supporting student decision-making.
2. To implement a weighted multi-criteria decision model.
3. To ensure robust input validation and semantic relevance checking.
4. To handle edge cases such as equal weights or tied option scores.
5. To generate transparent and explainable recommendations.
6. To maintain a fully dynamic system without static or hardcoded examples.

Scope of the Project

The system is designed to support decisions related to:

- Academic courses and majors
- Final-year or research project topics
- Internships and job offers
- Study vs work decisions
- University and postgraduate program selection

The bot is not intended for medical, legal, or financial decision-making beyond an academic context.

Literature Review and Related Work

Decision Support Systems (DSS) have been widely studied in fields such as operations research, management science and artificial intelligence. Traditional DSS models include:

- Analytic Hierarchy Process (AHP)
- Multi-Attribute Utility Theory (MAUT)
- Weighted Sum Models (WSM)

While these models are mathematically sound, their adoption among students remains limited due to:

- High cognitive overhead
- Lack of user-friendly interfaces
- Poor explainability

Recent advances in conversational AI and chatbots have enabled more natural user interaction. However, many educational chatbots focus on tutoring or information retrieval rather than decision reasoning. This project bridges the gap by combining conversational AI with formal decision models and explainability principles.

System Architecture

Overview

The system follows a modular, state-aware architecture consisting of:

1. Telegram User Interface
2. Conversation State Manager

3. Input Validation Engine
4. Decision Modeling Engine
5. Explanation Generator

Technology Stack

- Programming Language: Python
- Messaging Platform: Telegram
- Bot Framework: Telegram Bot API
- Decision Model: Weighted Scoring Model

Methodology

Decision Validation

The bot first validates that the user's problem belongs to a recognized student decision domain (e.g., career, project, course). Irrelevant or nonsensical inputs are rejected with gentle corrective feedback. The workflow does not proceed until a valid decision context is established.

Option Collection

Users are prompted to provide multiple decision options. Each option is checked for:

- Semantic relevance
- Duplication
- Completeness

A minimum of two valid options is enforced.

Criteria Definition and Weight Assignment

Users define decision criteria (e.g., difficulty, feasibility). Each criterion is assigned a weight on a 1–10 scale, where higher values indicate greater importance. The system validates numerical range and meaning.

Option Scoring

Each option is scored against each criterion using the same 1–10 scale. Invalid or non-numeric inputs are rejected, ensuring data integrity.

Weighted Decision Computation

The final score for each option is computed as:

$$\text{Final Score} = \sum (\text{Criterion Weight} \times \text{Option Score})$$

Options are ranked from highest to lowest score.

Edge-Case Handling

The system explicitly addresses common decision pitfalls:

- Equal Weights: Users are warned that no priority has been expressed and are encouraged to refine weights.
- Tied Scores: The bot avoids arbitrary recommendations and explains why options are equivalent, suggesting refinement of criteria or scores.

Explainability and Recommendation Logic

Rather than providing a simple ranking, the bot generates a natural-language explanation that:

- References the highest-weighted criteria
- Highlights strengths and weaknesses of top options

- Justifies the final recommendation logically

This ensures trust, transparency and learning for the user.

Implementation Details

The bot is implemented as a fully deployable Python application. Key design principles include:

- Modular function separation
- Stateful conversation management
- Defensive programming and validation
- No static or hardcoded examples

The system is extensible and suitable for future integration with large language models for semantic classification.

Testing and Results

The system was tested using multiple decision scenarios, including:

- Project topic selection
- Internship comparison
- Course selection

Results demonstrate that the bot:

- Prevents irrelevant or invalid interactions
- Produces consistent and rational rankings
- Provides clear explanations aligned with user priorities

Limitations

- The model relies on user-provided scores, which may still carry bias.
- Semantic validation is rule-based and may be improved with ML models.
- The system currently operates in English only.

Future Work

Potential extensions include:

- Integration of NLP-based intent classification
- Automatic suggestion of criteria based on decision type
- Visualization of decision matrices
- Web and mobile front-end versions

Conclusion

This project demonstrates the effectiveness of combining decision theory, conversational AI and explainable reasoning to address a real-world student problem. The AI Decision Recommendation Bot provides a structured, transparent and user-friendly approach to complex decision-making, empowering students to make informed choices with confidence.

The system successfully transforms vague decision confusion into rational, defensible recommendations, validating its relevance within educational and mentorship-driven AI development contexts.

